

Where does the water AI uses go?

Posted by u/Jrolaoni • 0 points • 21 comments

I mean it's not like it gets removed from the planet right? It will go back through the water cycle. Same as a nuclear power plant.

Comments

u/NameLips • 1 points • 2026-02-05 01:32 UTC

I think I see what you're getting at.

But there isn't an unlimited supply of water on Earth.

There are now rivers that no longer reach the sea, and entire seas (the Aral sea) have dried up, due to being overused for human agriculture. The notion of "water being infinite" doesn't mean they can build an infinite number of farms next to a river, does it? Because, obviously, that river can only supply a limited number of gallons per second. That river is a finite resource, despite being endlessly refilled.

Most aquifers took millions of years to fill, and we have drained them in a matter of decades.

The water cycle is like the rivers, except on a grand scale. It can only supply a certain, finite amount of fresh water every day. More and more of that very limited supply is being dragged into human use, which means it can't be used elsewhere.

Water rights are very sticky legal situations. Often you do not own the water rights to land you own. They're owned by the state and divvied up to agricultural and industrial interests. It's a big point of contention between states which share the same river, if one state uses up all the water, that means they're actively damaging agriculture in the state further downstream. They need to negotiate water use.

AI data centers are just another human industry sucking away at our limited supply. That means less for everybody else.

u/Optimoprime • 2 points • 2026-02-04 16:49 UTC

If I draw water from an underground aquifer, or a river, or a lake, and I evaporate it, that water doesn't go back into those bodies of water. The rain that falls from that evaporation is rain that may occur thousands of miles away.

It's kind of like when you spend money. The money doesn't disappear, but it also isn't yours anymore. You only have so much money in the bank to spend, and if you use more money than you make, eventually you'll run out.

u/KronusIV • 2 points • 2026-02-04 16:36 UTC

The data centers use evaporative cooling. They take drinking water in and shoot it into the atmosphere as vapor to take away heat. So yeah, it will eventually rain down and return to the water cycle. But that'll be hundreds or thousands of miles away. It's not being returned to the town where they took the water to start with.

u/newimprovedmoo • 1 points • 2026-02-04 17:57 UTC

And statistically the most likely place for it to fall is the ocean, where it will become basically useless to humans, as we require fresh water for most purposes.

u/DavyJonesCousinsDog • 1 points • 2026-02-04 16:31 UTC

Either the atmosphere or the local sewage system.

u/Delehal • 4 points • 2026-02-04 15:55 UTC

Most data centers rely on evaporative cooling. Meaning the computers are kept cool by heating up water and allowing that water to evaporate into the atmosphere. That water isn't destroyed. It goes back through the water cycle.

However, keep in mind that there is only so much fresh, usable water in the world at any given moment. There are constraints such as our ability to collect and purify water to make it safe for people to use. You may have heard of some places facing a drought when they have a shortage of usable water. There is always plenty of water around. The oceans are huge! But we can't drink ocean water without expensive desalination treatments.

u/luxivyn • 1 points • 2026-02-04 15:54 UTC

AI's water doesn't "go" anywhere. It gets so emotionally heated by our dumb questions it evaporates immediately

u/nomorehersky • 1 points • 2026-02-04 15:51 UTC

Most of it either evaporates during cooling or gets released back into local systems after being heated. Evaporated water eventually re enters the water cycle as rain.

u/noggin-scratcher • 2 points • 2026-02-04 15:51 UTC

There's no shortage of H₂O molecules on the planet: most of the surface is ocean. But most of the Earth's water is full of salt, or is difficult to access because it's deep underground, or at very least wouldn't be drinkable without a bit of treatment against pathogens and contaminants.

People and agriculture and data centres are all competing for a relatively limited supply of _fresh clean water_. It re-enters the water cycle after any of those uses, but isn't immediately available to use again - and there's only so much new supply from rivers to go around before we'd have to resort to expensive desalination.

u/Superior_Mirage • 1 points • 2026-02-05 06:22 UTC

Out of all the concerns AI raises, water usage is the one that nobody needs to care about.

Turf (lawns, golf courses, parks, etc.) uses 10-100x more water than AI (range is due to how you calculate AI water usage).

The only real concern with AI is if your municipality is stupid enough to authorize construction of a data center without the infrastructure to support it -- so don't elect stupid local officials, and you'll be fine.

u/-aVOIDant- • 1 points • 2026-02-04 16:07 UTC

Why do the data centers need fresh water for cooling?

u/noggin-scratcher • 2 points • 2026-02-04 16:16 UTC

Cooling systems don't work well when they're crusty with salt, and seawater can be surprisingly corrosive.

u/WiseAnimator7081 • 2 points • 2026-02-04 16:10 UTC

Salt water would corrode and therefore gum up the system. Not ideal of course.

u/Forest_Orc • 5 points • 2026-02-04 15:49 UTC

It evaporates and fall back as rain. However,

We don't know when/where it fall-back, and there is only so many water which can be pumped from the ground at a given time. I am sure you've seen some dry season restriction like *forbiden to water your garden or wash your car* so we keep water for farmer and city-water pump. Now we have data-centre using AI too

u/bangbangracer • 4 points • 2026-02-04 15:48 UTC

Grey water and evaporation.

The water that comes out needs to be treated as grey water before it can go back to the reservoir, and a large amount is lost to evaporation.

Why is this a problem? The water cycle isn't very fast and the AI data centers draw more water that the infrastructure can usually handle and faster than the reservoirs can replenish through the water cycle.

u/KronusIV • 1 points • 2026-02-04 16:39 UTC

It isn't exactly "lost" to evaporation, evaporation is the point. There is no gray water here, the water is sent into the atmosphere, taking heat away with it.

u/Jrolaoni • 1 points • 2026-02-04 15:50 UTC

Do nuclear plants do the same thing?

u/bangbangracer • 2 points • 2026-02-04 15:52 UTC

Yes and no.

Nuke plants do pretty much the same thing, but they actually have to have capture and treatment facilities on site. That's to prevent accidental contamination. They also run the water through multiple times.

u/Jrolaoni • 1 points • 2026-02-04 15:56 UTC

Okay, thanks. Your answers seem the most thorough.

u/bangbangracer • 1 points • 2026-02-04 16:01 UTC

Thank you.

We also have to bring up the concept of utils and utility. Utils are an economic concept of the utility that can be extracted from something.

A nuke plant is producing so many utils by making gigawatts of electricity. The utility benefit massively outweighs the costs. Meanwhile, to make one AI image, it can take about the same amount of electricity as running a 1000w microwave for 3-5 hours.

Relative to resources consumed and what is being made, AI is not proving it's utility.

u/Sayakai • 4 points • 2026-02-04 15:51 UTC

Yes.

But in exchange we get a Gigawatt worth of power instead of AI slop.

u/bangbangracer • 3 points • 2026-02-04 15:55 UTC

I forgot to mention this aspect as well. We need to consider utility. Nuke plants produce massive amounts of electricity relative to their water consumption. One AI image is close to 3-5 hours of microwave usage. Relative to the energy consumed and water used, we aren't getting that much utility.